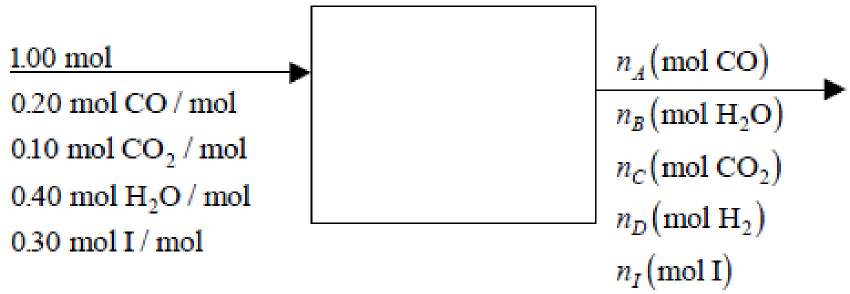
a. $CO + H_2O \longleftrightarrow CO_2 + H_2$

(A) (B) (C) (D)



Degree of freedom analysis: 6 unknowns $(n_A, n_B, n_C, n_D, n_I, \xi)$

$-$ 4 expressions for $n_i(\xi)$

$-$ 1 balance on I

$-$ 1 equilibrium relationship

0 DF

b. Since two moles are prodcued for every two moles that react,

$$(n_{\text{total}})_{\text{out}} = (n_{\text{total}})_{\text{in}} = \underline{\underline{1.00(\text{mol})}}$$

$$n_A = 0.20 - \xi \qquad (1)$$

$$n_B = 0.40 - \xi \qquad (2)$$

$$n_C = 0.10 + \xi \qquad (3)$$

$$n_D = \xi \qquad (4)$$

$$n_I = 0.30 \qquad (5)$$

$$n_{tot} = 1.00\,\text{mol}$$

At equilibrium: $\dfrac{y_C y_D}{y_A y_B} = \dfrac{n_C n_D}{n_A n_B} = \dfrac{(0.10+\xi)(\xi)}{(0.20-\xi)(0.40-\xi)} = 0.0247\exp\left(\dfrac{4020}{1123}\right) \Rightarrow \xi = 0.110 \text{ mol}$

$$y_D = n_D = \xi = \underline{\underline{0.110(\text{mol}\,H_2/\text{mol})}}$$

c. The reaction has not reached equilibrium yet.

d.

T (K)	x (CO)	x (H_2O)	x (CO_2)	Keq	Keq (Goal Seek)	Extent of Reaction	y (H_2)
1223	0.5	0.5	0	0.6610	0.6610	0.2242	0.224
1123	0.5	0.5	0	0.8858	0.8856	0.2424	0.242
1023	0.5	0.5	0	1.2569	1.2569	0.2643	0.264
923	0.5	0.5	0	1.9240	1.9242	0.2905	0.291
823	0.5	0.5	0	3.2662	3.2661	0.3219	0.322
723	0.5	0.5	0	6.4187	6.4188	0.3585	0.358
623	0.5	0.5	0	15.6692	15.6692	0.3992	0.399
673	0.5	0.5	0	9.7017	9.7011	0.3785	0.378
698	0.5	0.5	0	7.8331	7.8331	0.3684	0.368
688	0.5	0.5	0	8.5171	8.5177	0.3724	0.372
1123	0.2	0.4	0.1	0.8858	0.8863	0.1101	0.110
1123	0.4	0.2	0.1	0.8858	0.8857	0.1100	0.110
1123	0.3	0.3	0	0.8858	0.8856	0.1454	0.145
1123	0.5	0.4	0	0.8858	0.8867	0.2156	0.216

The lower the temperature, the higher the extent of reaction. An equimolar feed ratio of carbon monoxide and water also maximizes the extent of reaction.